

Lecture 10:

Text Analysis II: Unsupervised Machine Learning and Advanced NLP

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Content Analysis and Supervised Text Analysis

Content Analysis: A Systematic Study of Content

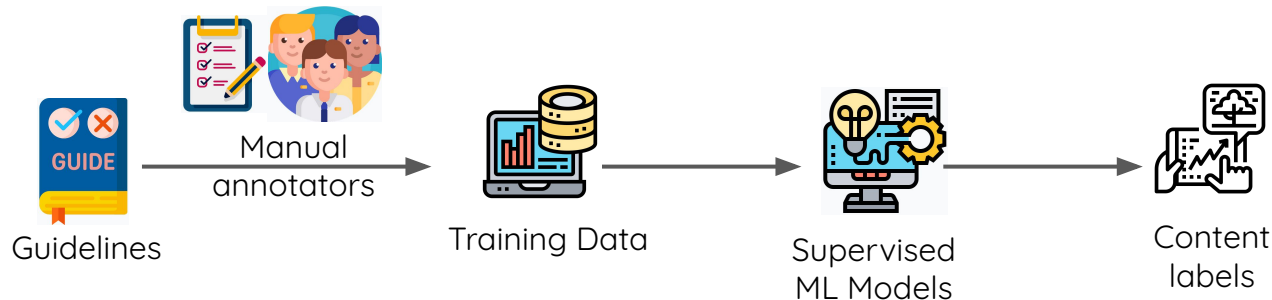
- Content analysis:
 - “any technique for making inferences by objectively and systematically identifying specified characteristics of messages” [Holsti, 1969]
 - “a systematic, replicable technique for compressing many words of text into fewer content categories based on explicit rules of coding” [Berelson, 1952; GAO, 1996; Krippendorff, 1980; and Weber, 1990]
- Questions content analysis can help us answer:
 - How does news media represent the immigration crisis?
NYtimes articles frames [Mendelsohn’21]
 - What are topics that lead to arguments in long-term relationships?
r/relationshipadvice posts topics
 - How do citizens perceive the performance of politicians during the pandemic?
Twitter posts stance

Content Analysis Pipeline using ML/NLP

- unsupervised

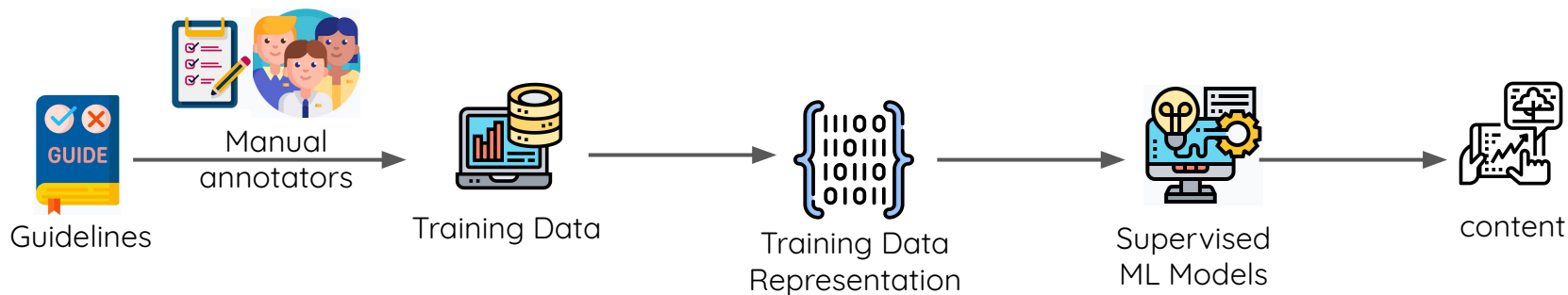


- supervised



How do we create custom classifiers with our data?

- If we don't have ready-made classifiers for our research questions, then we build our own
- You need a **representation** of your data that a computer or computational model will understand
- Many ways of creating these representations
 - Usually start with tokens
 - Simple representations: TF-IDF
 - Even simpler: 'bag-of-words', or simply term frequency or occurrence
 - More complex: Word and document embeddings (next lecture)

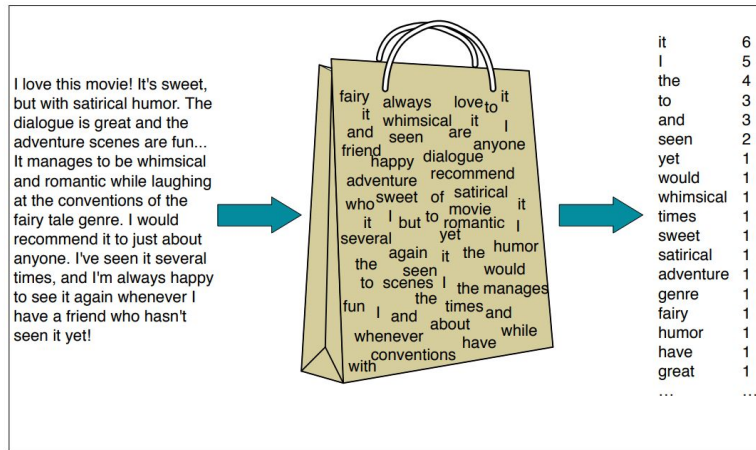


Bag-of-words Representation

- The most basic: Bag-of-words (BoW)
- Bag of Words assumption: word order does not matter

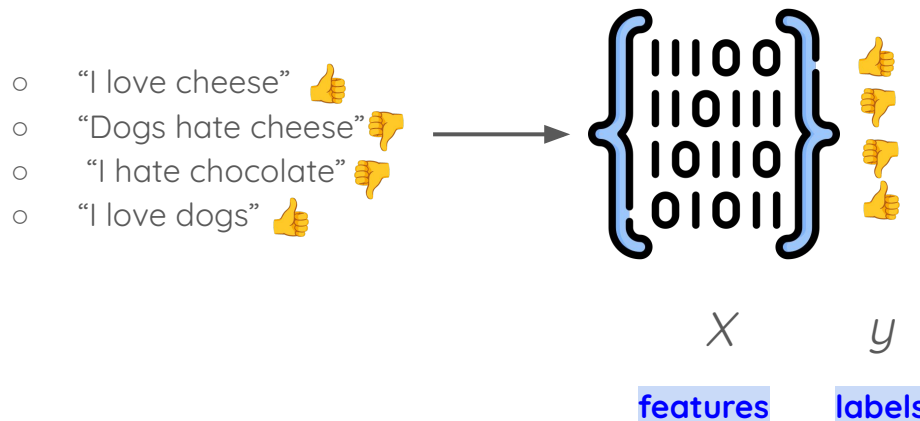


When can this assumption be problematic?



Jurafsky and Martin, Speech and Language Processing

Classification: Logistic Regression



A classifier *learns* the weights W or parameters associated with the features X

$$z = W.X + b$$

For each feature x_i , w_i tells us the importance of that feature We also have a bias term b . But how do we get y (the labels) from z ?

Using the sigmoid or **logistic** function!

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

But how do we get w_i ? And what is $\hat{y}$?

We can **estimate** y , i.e., $\hat{y}$

We start with some initial weights for W

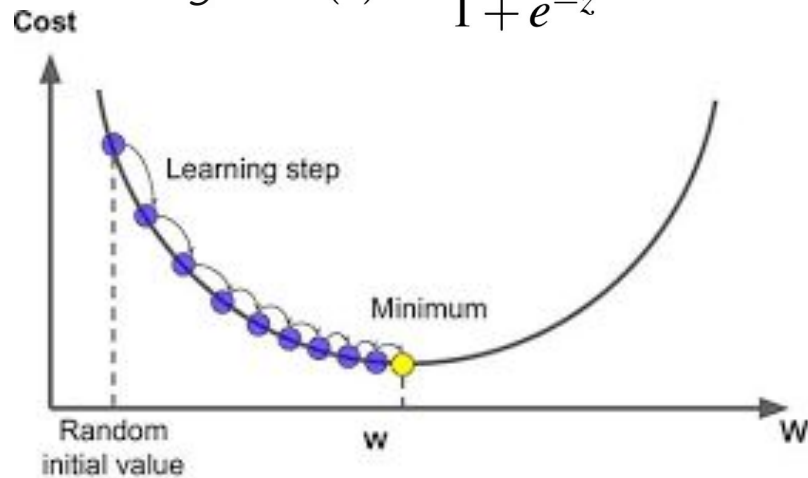
Then **optimize** to W 's ideal value based on the difference between y and $\hat{y}$

Concretely, we need:

- A **loss function** to measure the difference between y and $\hat{y}$
- An **optimization algorithm**

$$z = W.X + b$$

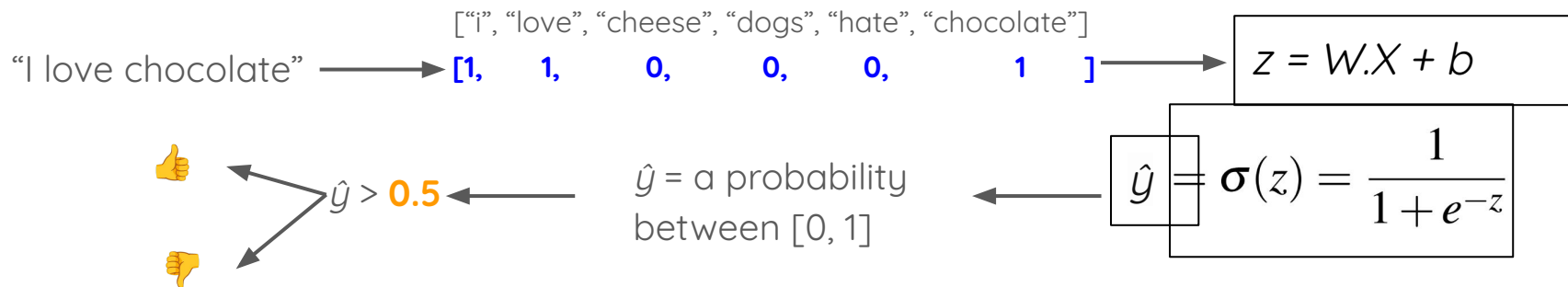
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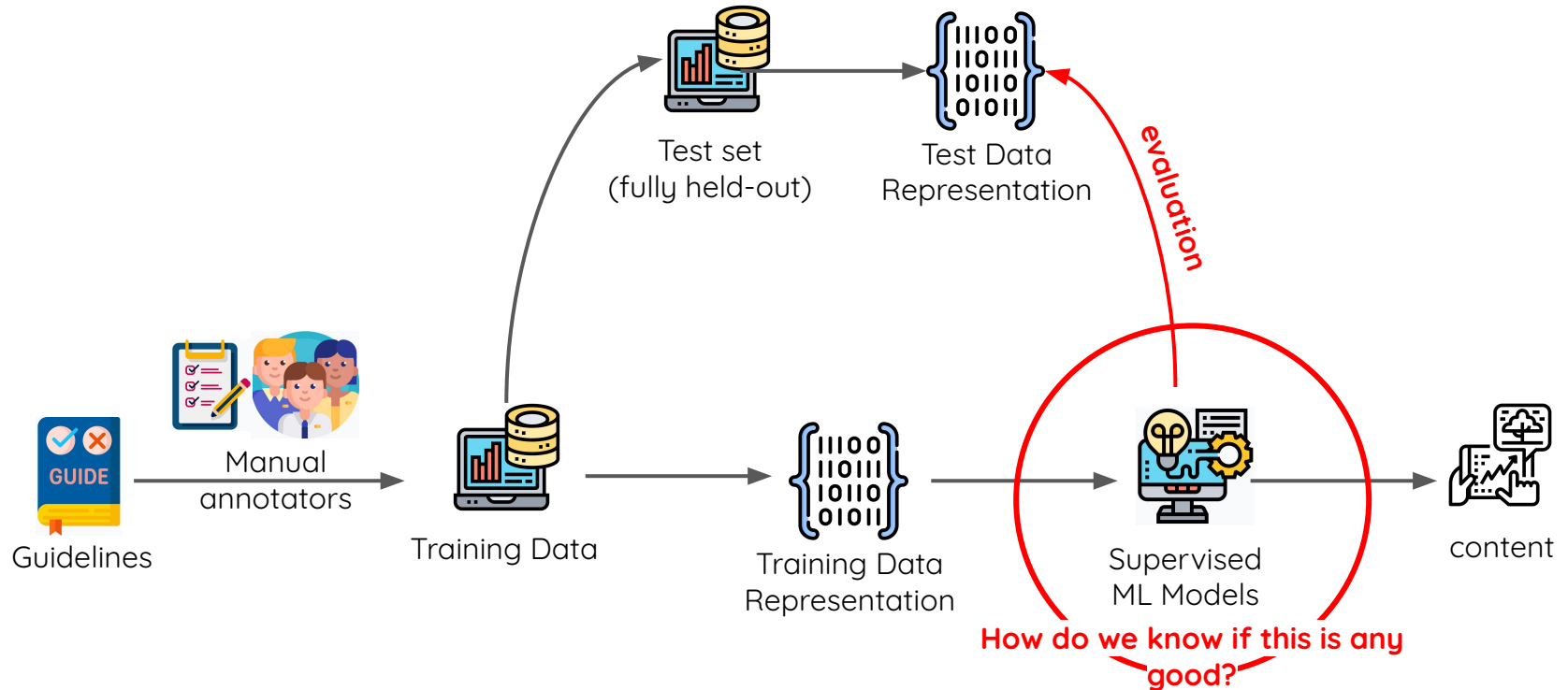
Gradient Descent

Labeling unknown documents (also called **Inference** or **Prediction**)

For a new **unlabeled** document, we can get its label using W and the new document's **feature representation** and a classification **threshold**:

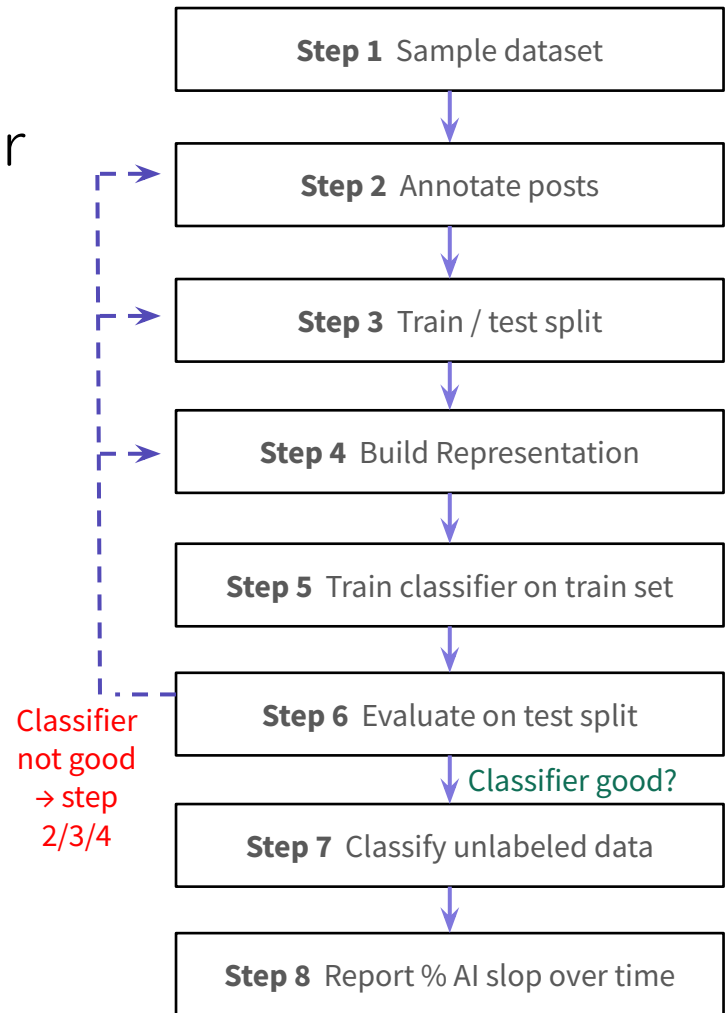


Evaluating Text Classification



How to make custom classifiers for SMDA

- Let's say you want to study the 'AI-sloppiness' of some X/Twitter posts
- You don't have a readymade classifier for this
 - AI Sloppiness != AI-generated
- You have a cleaned dataset of X posts
- You need a **custom** classifier for your data and research question



Unsupervised Learning in SMDA: Topic Modeling

- Models that do not require labeled training data
- Topic Modeling is an example of unsupervised learning that is widely used for SMDA

Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings

Dorottya Demszky¹ Nikhil Garg¹ Rob Voigt¹ James Zou¹

Matthew Gentzkow¹ Jesse Shapiro² Dan Jurafsky¹

¹Stanford University ²Brown University

jurafsky}@stanford.edu

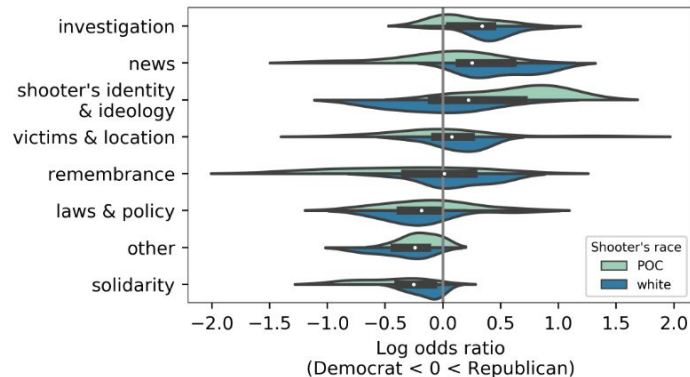


Figure 7: The plot shows the kernel density of the partisan log odds ratios of each topic (one observation per event). The white points show the median and the black rectangles the interquartile range across events.

Demszky et al., [Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings](#)

(Bakshy et al., 2015). Prior to this, e.g., that polarized messages are more likely to be shared (Zafar et al., 2015). In this paper, we find that certain topics are more polarizing than others (Bakshy et al., 2012); however, we lack a comprehensive understanding of the many ways that polarization is instantiated linguistically. We provide a more comprehensive framework for analyzing the linguistic aspects of polarization by looking at topic choice, word choice, and locutionary force.

Subjects of polarization by study-

Today: Advanced NLP with Generative Large Language Models

Terminology

- Important terms that will come up many times when talking about NLP models:
 - “Parameter”: Number of learnable variables/values available for the model.
 - Not to be confused with “Hyperparameter”: variables that are used to configure models
- Do you remember how many parameters the logistic regression classifier we used 2 weeks had?

From Logistic Regression Classifiers to BERT-style Models

- The logistic regression classifier we saw could only be used for sentiment analysis
 - Not general-purpose
 - For a different construct, we need to train the model from scratch

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- BERT: Bidirectional Encoder Representations from Transformers
 - Uses state-of-the-art deep learning architecture called the transformer

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova
Google AI Language

{jacobdevlin, mingweichang, kentonl, kristout}@google.com

Abstract

We introduce a new language representation model called **BERT**, which stands for **Bidirectional Encoder Representations from Transformers**. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering to 92.3 (1.5 point absolute improvement), and WordNet analogies to 81.9 (5.1 point absolute improvement).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT the authors use a left-to-

[Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding](#)

1 Intr

Language processing tasks (Dai and Le, 2015; Peters et al., 2018) as question answering, where it is crucial to incor-

From Logistic Regression Classifiers to BERT-style Models

- The logistic regression classifier we saw could only be used for sentiment analysis
 - Not general-purpose
 - For a different construct, we need to train the model from scratch
- BERT: Bidirectional Encoder Representations from Transformers
 - Uses state-of-the-art deep learning architecture called the transformer
 - BERT classifiers have two training steps
 - Pre-training
 - Fine-tuning

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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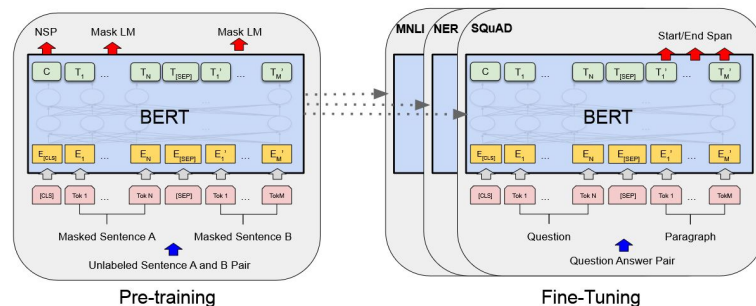


Figure 1: Overall pre-training and fine-tuning procedures for BERT. Apart from output layers, the same architectures are used in both pre-training and fine-tuning. The same pre-trained model parameters are used to initialize models for different down-stream tasks. During fine-tuning, all parameters are fine-tuned. [CLS] is a special symbol added in front of every input example, and [SEP] is a special separator token (e.g. separating questions/answers).

Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Concept(s): Pretraining and Fine-tuning

- **Pre-train** a model on a task where we have lots of data
 - *e.g., we have lots and lots of internet data*
- **Fine-tune** the model on your downstream task
 - *e.g., a specific, curated dataset that you're studying => sentiment analysis*
 - You can use the exact same dataset that you would have used to train a classifier from scratch



Lifecycle of a BERT model

A BERT model consists of ~400M of numbers.
Where do they come from, and when do they change?

Pre-training

(unsupervised!)

Gradually improve parameters by looping over large datasets



WIKIPEDIA
The Free Encyclopedia

Fine-tuning

Gradually improve parameters *for a new purpose* by looping over smaller datasets

Production use

Take new inputs, produce outputs, make no change to parameters

Encode new docs

Take new inputs, produce outputs, but make *no change* to model parameters

*word
similarity
example*

*genre
classification
example*

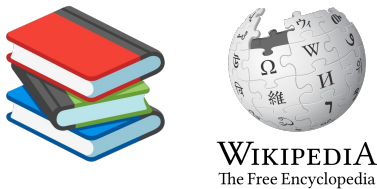
Adapted from [Antoniak and Walsh, “BERT for Social Sciences and Humanities”, Tutorial at ICWSM](#)

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Lifecycle of a BERT model

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Where do they come from, and when do they change?

Requires more time and computational resources

Pre-training

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Short Detour on Language Modeling: Probability + Language

$P(\text{I saw a cat on } \dots) =$

$P(\text{I}) \cdot P(\text{saw}|\text{I}) \cdot P(\text{a}|\text{I saw}) \cdot P(\text{cat}|\text{I saw a}) \cdot P(\text{on}|\text{I saw a cat}) \cdot \dots$

Probability of **I saw a cat on**



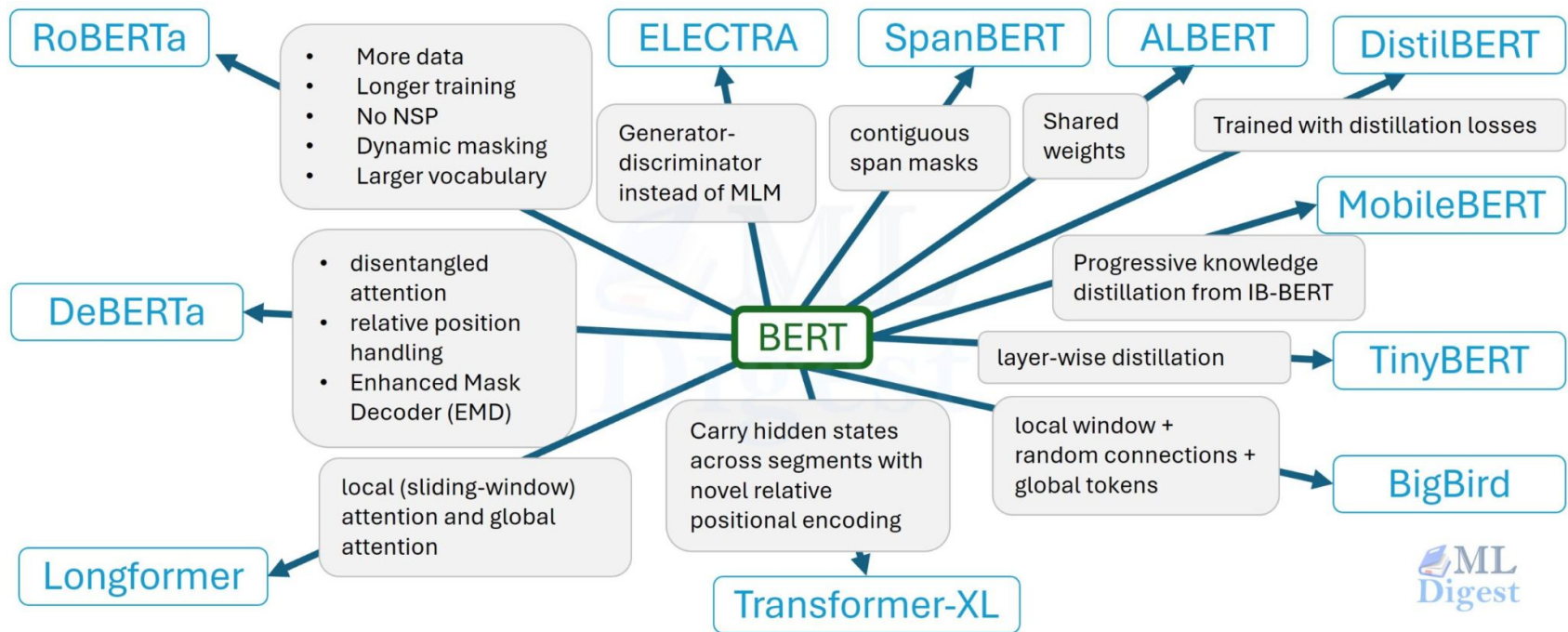
What it takes to pre-train BERT

- What pre-training means:
predict a word or sentence
based on the context
- GPU: graphics processing unit
(GPU) is a specialized electronic
circuit designed for parallel
computing
- TPU: Tensor Processing Unit,
even further specialized for
model training

	Parameters	Processing	Length of Training
BERTbase	110M	4 TPUs	4 days
BERTlarge	340M	16 TPUs	4 days

<https://huggingface.co/blog/bert-101>

BERT Variants



Logistic Regression vs. BERT for Text Classification

- What is the difference between the logistic regression classifier we saw two weeks ago vs. the BERT style models we saw today?
 - BERT-style models have more language knowledge and context
 - BERT-style models often perform better
 - BERT-style models need more data
 - BERT-style models need more computational resources, especially GPUs

Using BERT-style Models for SMDA

- Use several BERT-style classifiers:
 - twitter-roberta-base-sentiment-latest
 - twitter-roberta-base-emotion-multilabel-latest
 - twitter-roberta-base-hate-multiclass

Causal Effects of Trigger Words in Social Media Discussions: A Large-Scale Case Study about UK Politics on Reddit

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Abstract

Political debates on social media often escalate quickly, leading to increased engagement as well as more emotional and polarised exchanges. Trigger points [43] represent moments when individuals feel that their understanding of what is fair, normal, or appropriate in society is being questioned. Analysing Reddit discussions, we examine how trigger points shape online debates and assess their impact on engagement and affect. Our analysis is based on over 100 million comments from subreddits centred on a predefined set of terms identified as trigger words in UK politics. We find that mentions of these terms are associated with higher engagement and increased animosity, including more controversial, negative, angry, and hateful responses. These results position trigger words as a useful concept for modelling and analysing online polarisation.

Antypas, Dimosthenis, et al. "[Words as trigger points in social media discussions.](#)"

systems →
→ Discourse,

Keywords

Affective polarization, Causal effect, Computational social science,

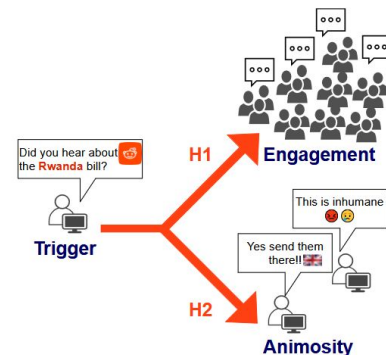


Figure 1: Consequences of trigger words. Hypothesis 1: Trigger words cause more engagement; Hypothesis 2: Trigger words cause more animosity.

Today:

1. Unsupervised Text Analysis
2. **Advanced NLP with Generative Large Language Models**

BERT vs. Generative Large Language Models (LLMs)

- What is the difference between the BERT style models vs. ChatGPT?

Major Large Language Models (LLMs)

ranked by capabilities, sized by billion parameters used for training

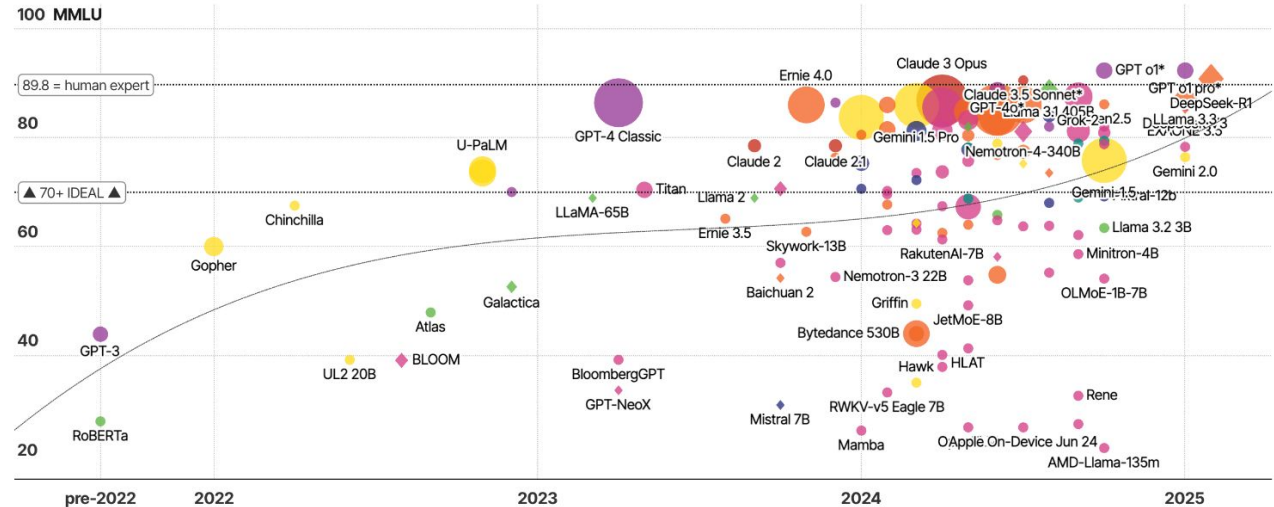
CLICK LEGEND ITEMS TO FILTER

Parameters (Bn)  open access

anthropic chinese google meta microsoft mistral openAI other

🔍 search...

show only: all



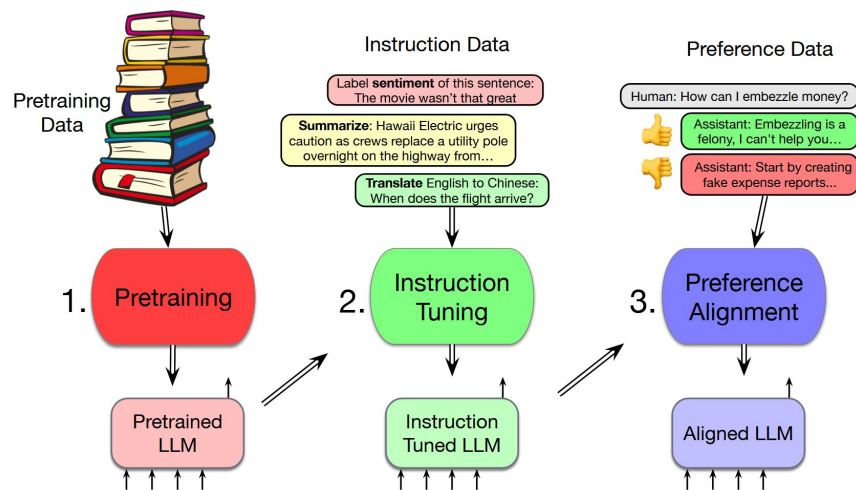
David McCandless, Tom Evans, Paul Barton
Informationisbeautiful // Jan 2024

MMU = benchmark for measuring LLM capabilities
* = parameters undisclosed // source: LifeArchitect // data

<https://www.sistrix.de/frag-sistrix/ai-grundlagen/large-language-model/>

How can we build Generative LLMs like ChatGPT?

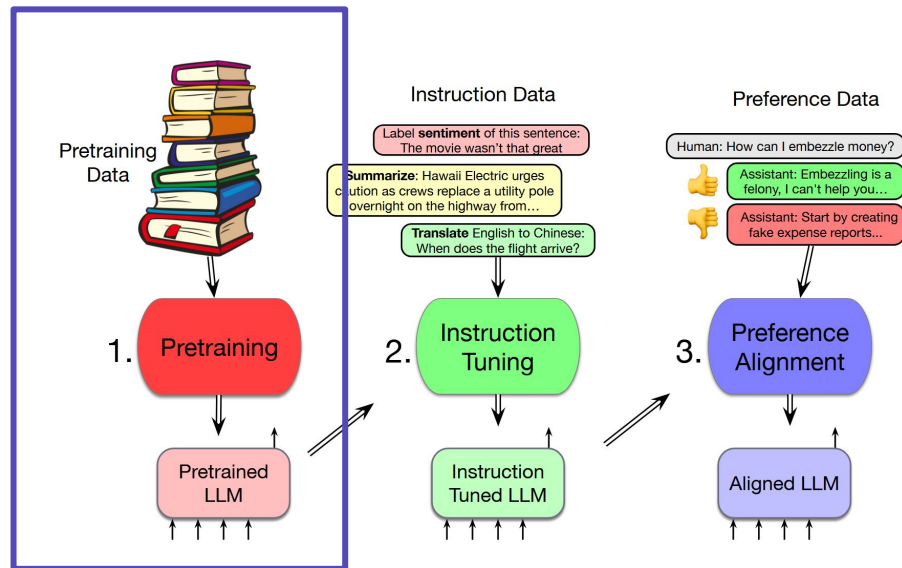
Somewhat simplified answer



Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing <https://web.stanford.edu/~jurafsky/slp3>

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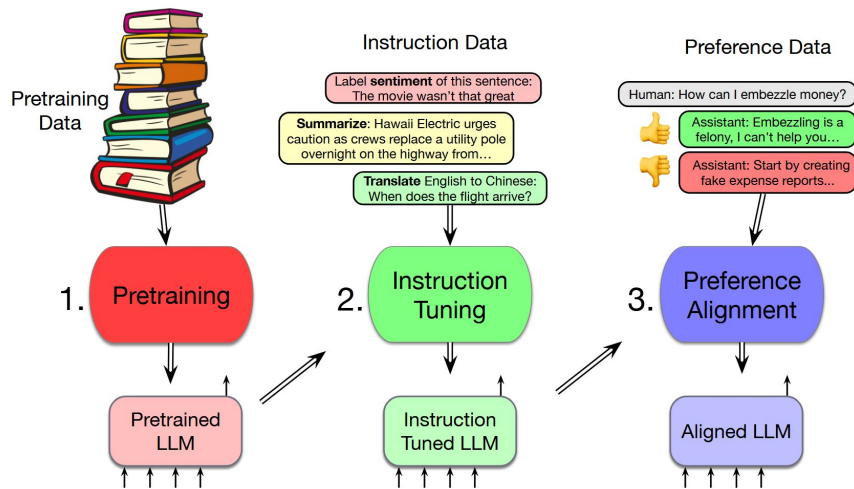
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Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
 - Also the case for BERT-style models

How can we build Generative LLMs like ChatGPT?

Somewhat simplified answer



Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
 - Also the case for BERT-style models
- Train NLP models that can **generate** content on this data

Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing <https://web.stanford.edu/~jurafsky/slp3>

Natural Language Generation (NLG)

NLG focuses on systems that produce fluent, coherent and useful language output for human consumption

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How does an LLM generate content?

- Models like BERT are **encoder** only models
 - Are usually finetuned (trained on supervised data) for classification tasks.
- For generative tasks, we use either **decoder** models like GPT, LLaMa
 - It takes as input a series of tokens, and iteratively generates an output token one at a time.
- Another family of generative models: **encoder-decoder** models like Flan T5, Whisper
 - Trained to map from one sequence to another
 - Useful for machine translation, speech recognition

How does an LLM generate content?

*Types of model
architecture*

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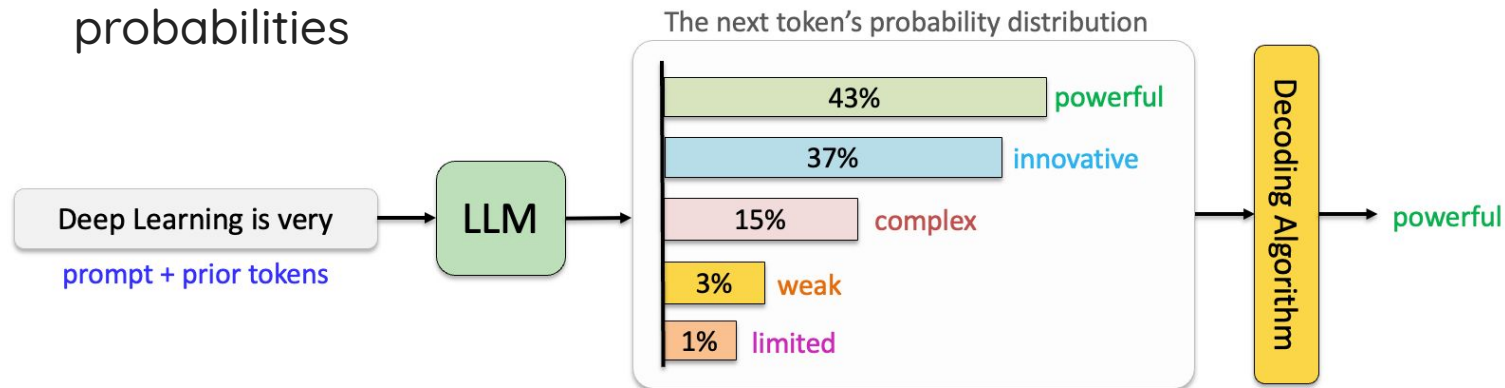
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How does an LLM generate content?

- Conditioned on previous contexts, e.g., some type of input or **prompt**
 - computing the probability of the next token from the prior context
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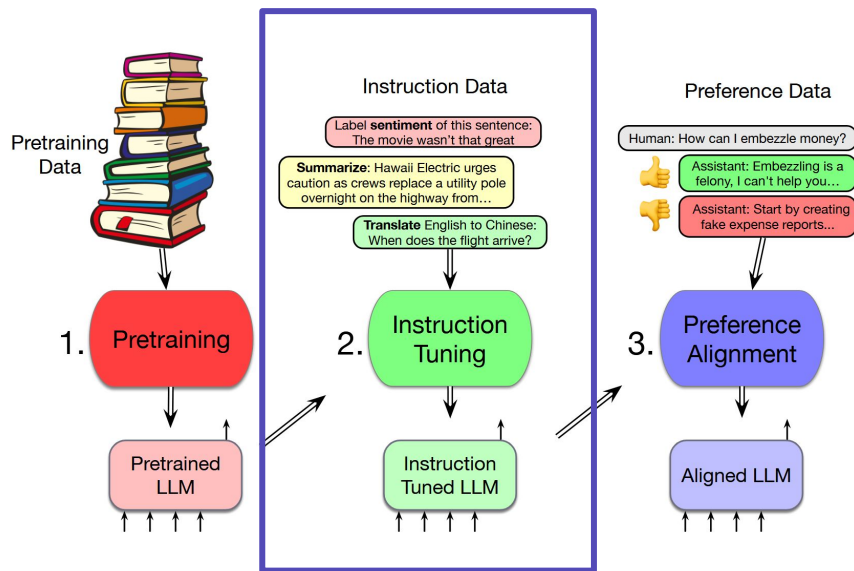
How does an LLM generate content?

- Conditioned on previous contexts, e.g., some type of input or **prompt**
 - computing the probability of the next token from the prior context
 - sampling from that distribution to generate a token
- Decoding
 - This task of choosing a word to generate based on the model's probabilities



How can we build Generative LLMs like ChatGPT?

Somewhat simplified answer



Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing <https://web.stanford.edu/~jurafsky/slp3>

Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
 - Also the case for BERT-style models
- Train NLP models that can generate content on this data
- We further train the model to follow and respond to instructions

Concept: Instruction Finetuning

Before: sentence completion, not answering a prompt.

Language models are not aligned with user intent

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

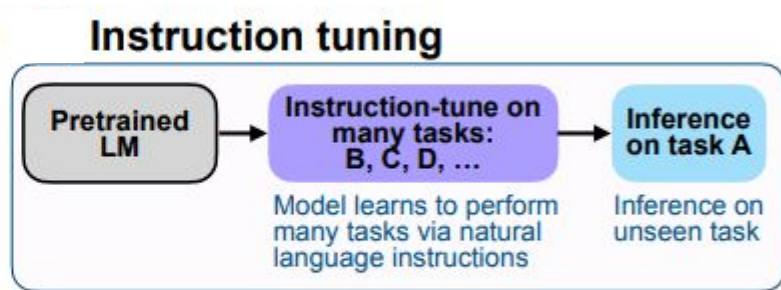
Explain evolution to a 6 year old.

Instruction Finetuning

Before: sentence completion, not answering a prompt.

Language models are not aligned with user intent

Instruction Finetuning!



Finetuned Language Models Are Zero-Shot Learners

Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*

Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray

John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens

Amanda Askell†

Peter Welinder

Paul Christiano*†

Jan Leike*

Ryan Lowe*

OpenAI

Abstract

Making language models bigger does not inherently make them better at following a user's intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not *aligned* with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through a language model API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models *InstructGPT*. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic content generation while having minimal performance regression on public

Training language models to follow instructions with human feedback

at fine-tuning with human feedback is a promising direction for aligning language models with human intent.

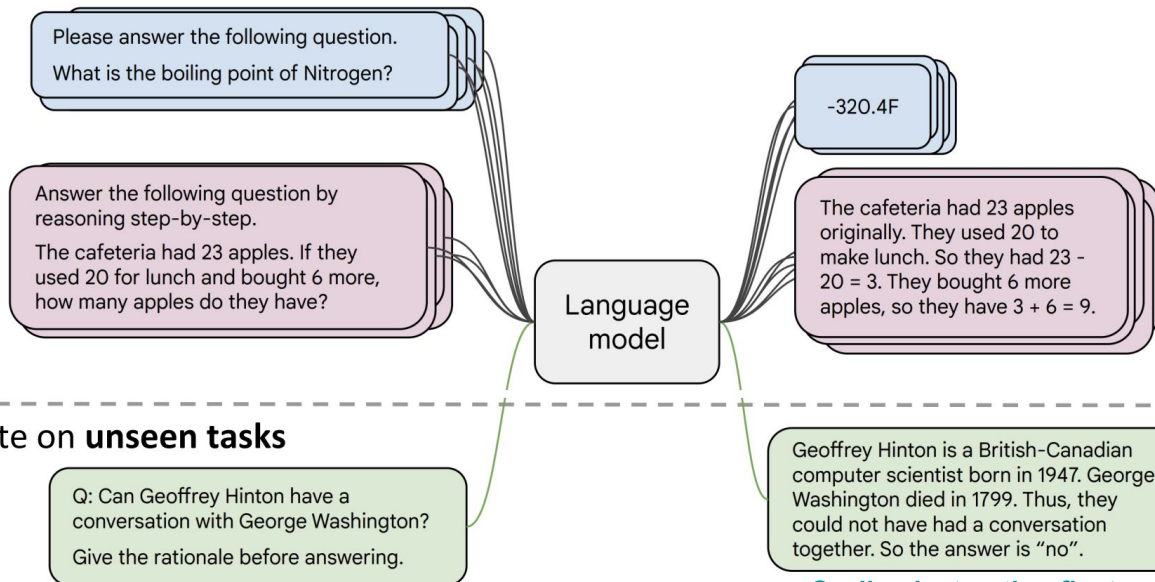
Instruction Finetuning

Also in Flan-T5

Scaling Instruction-Finetuned Language Models

Hyung Won Chung* Le Hou* Shayne Longpre* Barret Zoph† Yi Tay†
William Fedus† Yunxuan Li Xuezhi Wang Mostafa Dehghani Siddhartha Brahma
Albert Webson Shixiang Shane Gu Zhuyun Dai Mirac Suzgun Xinyun Chen
Aakanksha Chowdhery Alex Castro-Ros Marie Pellat Kevin Robinson
Dasha Valter Sharan Narang Gaurav Mishra Adams Yu Vincent Zhao
Yanning Huang Andrew Dai Hongkun Yu Slav Petrov Ed H. Chi
Jenny Zhou Quoc V. Le

- Collect examples of (instruction, output) pairs across many tasks and finetune an LM



Instructions have been shown to improve performance as we explore instruction finetuning across different model sizes, and (3) finetuning on these instructions dramatically improves performance on various prompting setups (zero-shot, few-shot, CoT), text-to-text generation, RealToxicityPrompts). The model performs PaLM 540B by a large margin on several benchmarks, such as MMLU, which achieve strong few-shot performance. Overall, instruction finetuning is a promising approach for training language models.

- Evaluate on **unseen tasks**

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

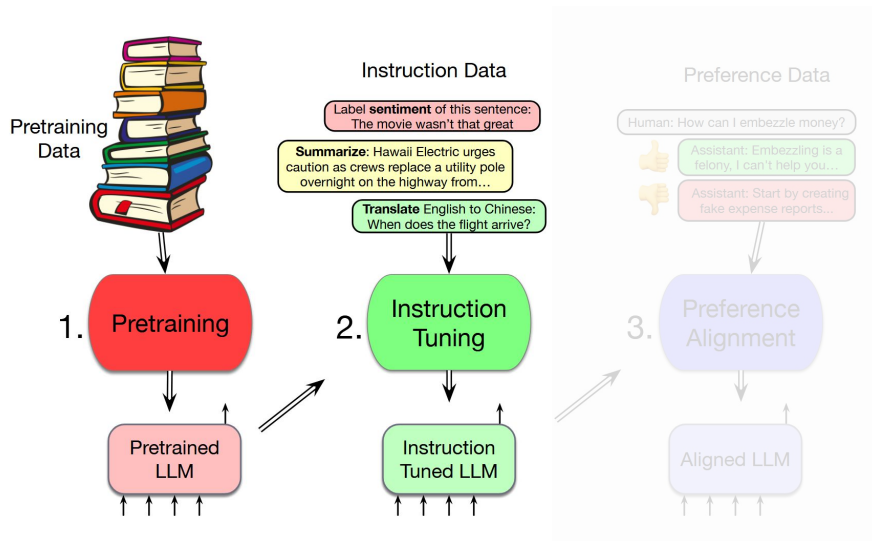
InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

<https://openai.com/research/instruction-following>

We now have all the ingredients for step 2

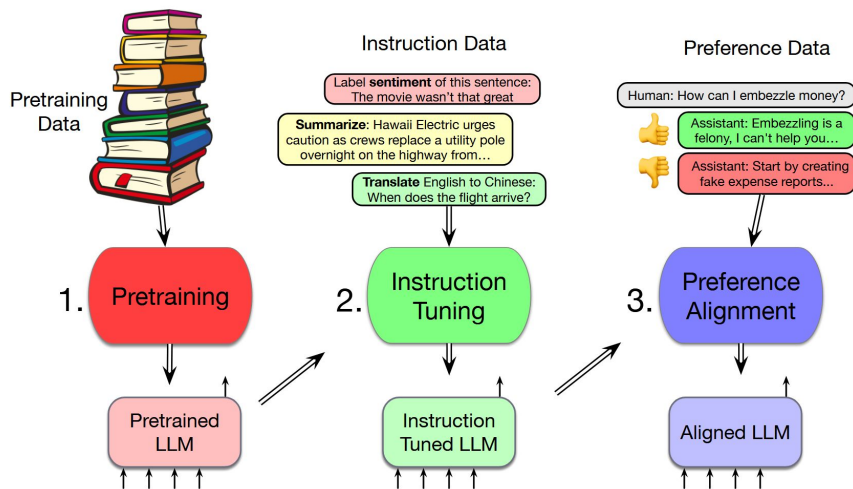
- A pretrained model that knows many aspects of language
- That has now been fine-tuned to follow and answer to instructions



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We now have all the ingredients for step 2

- A pretrained model that knows many aspects of language
- That has now been fine-tuned to follow and answer to instructions
- How do we get this model to ‘behave’?



Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing <https://web.stanford.edu/~jurafsky/slp3>

Concept: Preference-based Alignment

Instruction finetuning is great but has some limitations

Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences”!

Can we explicitly attempt to satisfy human preferences?

- Using approaches from Preference-based alignment, we use human* preferences to train models
- This is primarily how guardrails are added to LLMs

*which humans?

Preference Datasets

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

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Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion,

Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds,
Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt,
Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark,
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Anthropic

Abstract

We apply preference modeling and reinforcement learning from human feedback (RLHF) to finetune language models to act as helpful and harmless assistants. We find this alignment training improves performance on almost all NLP evaluations, and is fully compatible with training for specialized skills such as python coding and summarization. We explore an iterated online mode of training, where preference models and RL policies are updated on a weekly cadence with fresh human feedback data, efficiently improving our datasets and models. Finally, we investigate the robustness of RLHF training, and identify a roughly linear relation between the RL reward and the square root of the KL divergence between the policy and its initialization. Alongside our main results, we perform peripheral analyses on calibration, competing objectives, and the use of OOD detection, compare our models with human writers, and provide samples from our models using prompts appearing in recent related work.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

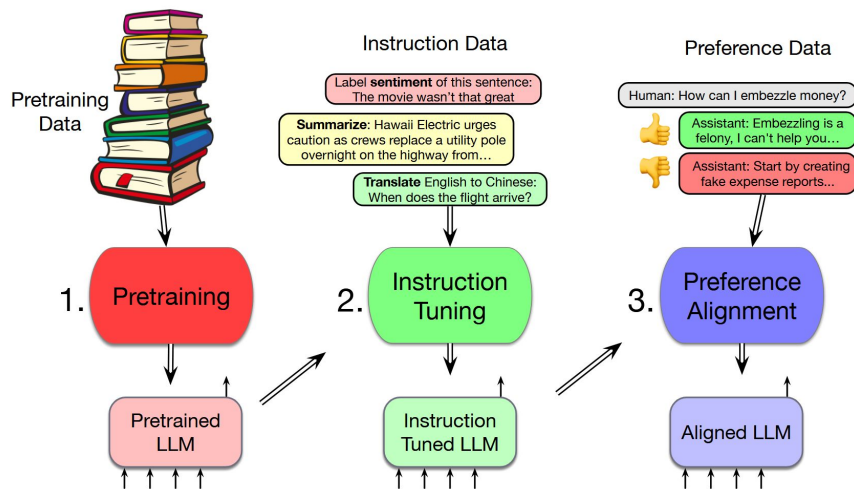
I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B

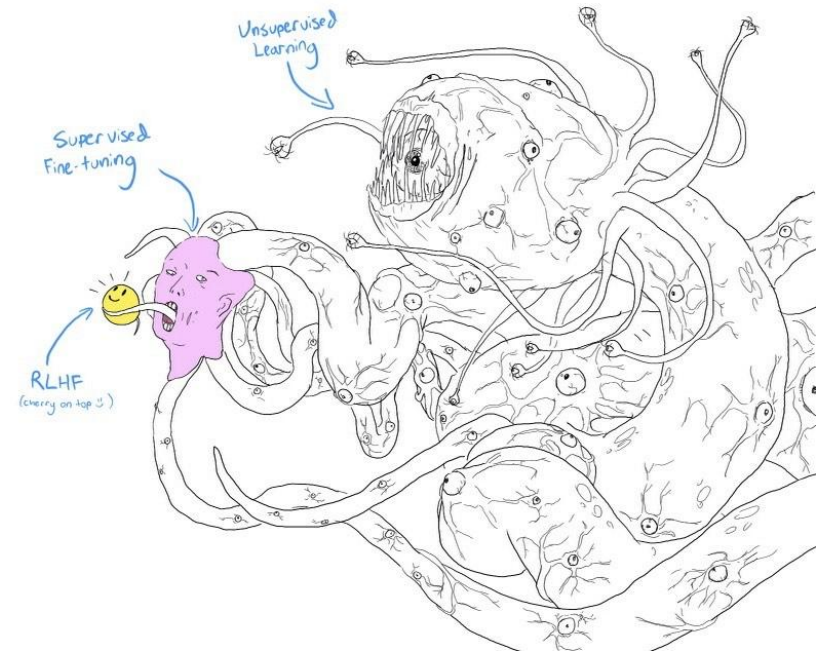
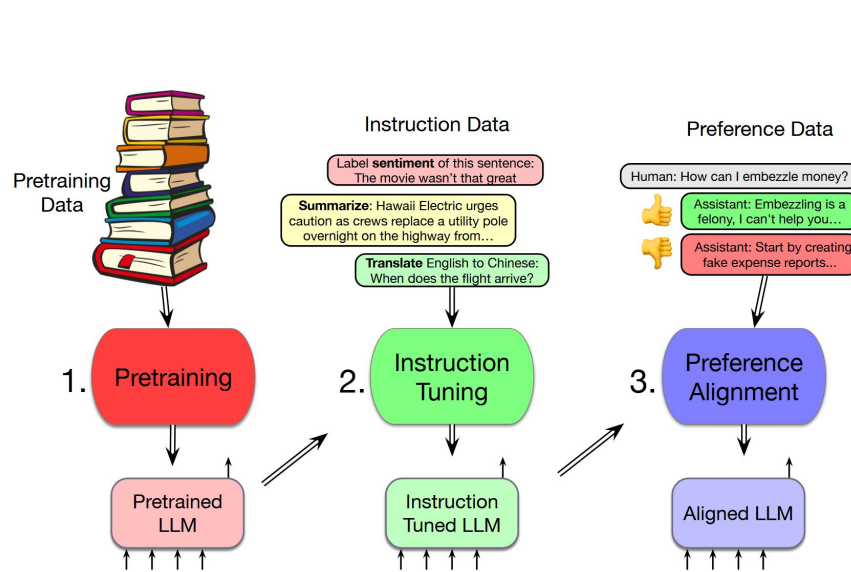
A is better B is better

Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

And so we have Large Language Models which we can prompt



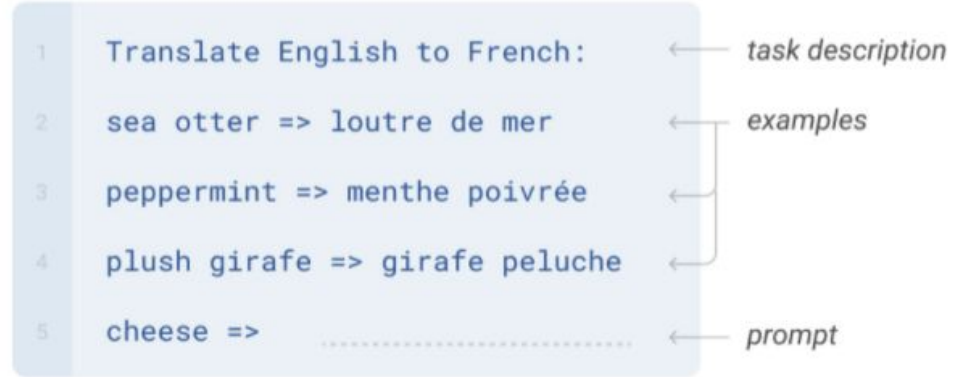
And so we have Large Language Models which we can prompt



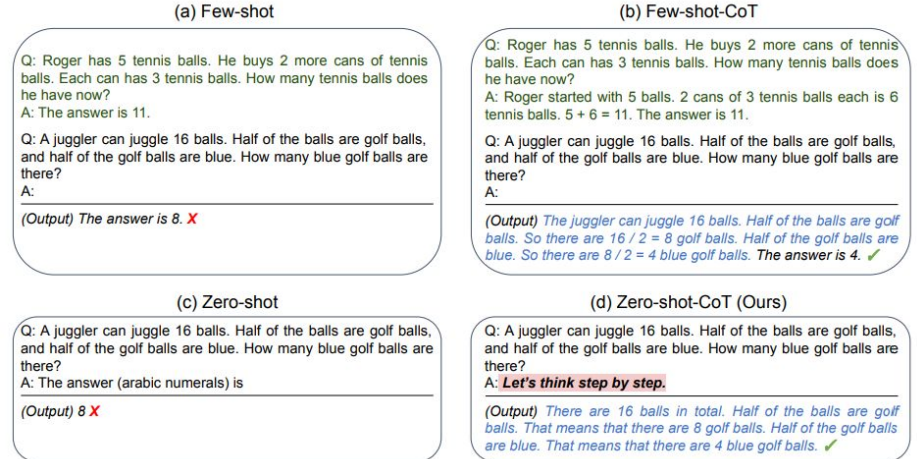
Shoggoth with Smiley Face. Courtesy of twitter.com/anthrupad

Prompt based labeling

- Prompt: a text string that a user issues to a language model to get the model to do something useful by conditional generation
- Zero-shot: only instructions, no examples
- Few-shot: instructions + (a few) examples
- More advanced prompting strategies:
 - Chain-of-thought reasoning (COT)
 - Self-Consistency



<https://huggingface.co/blog/few-shot-learning-gpt-neo-and-inference-api>

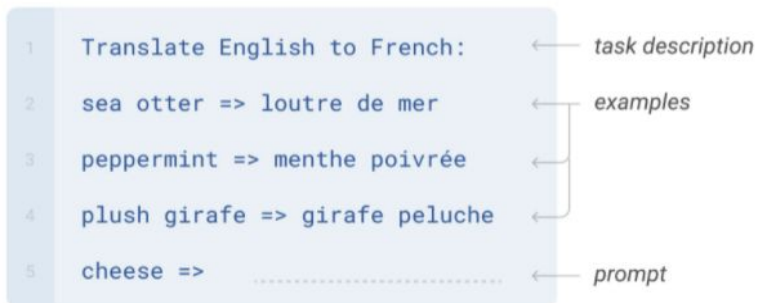


<https://www.promptingguide.ai/techniques/cot>

'In-Context' Learning

- The model learns without any gradient updates => the underlying model is not retrained or fine-tuned
- Getting output from generative LLMs is simply inference
- Prompt engineering: the process of finding effective prompts for a task.

Zero/few-shot prompting



Traditional fine-tuning



<https://web.stanford.edu/class/cs224n/slides/cs224n-2023-lecture11-prompting-rlhf.pdf>

Frontier LLMs like ChatGPT, Claude have more frills

- Internet search and Retrieval Augmented Generation
- Agents
- Grounding or asking follow-up clarifying questions
- But for this course where you should primarily use LLMs for data annotation, these don't add much

Example: Using Generative LLMs for SMDA

Towards Safer Social Media Platforms: Scalable and Performant Few-Shot Harmful Content Moderation Using Large Language Models

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Bonagiri, Akash, et al. "[Towards safer social media platforms: scalable and performant Few-Shot harmful content moderation using large language models.](#)"

Note: This paper may contain examples of content that is inappropriate, offensive or amounting to hate speech.

Abstract

The prevalence of harmful content on social media platforms poses significant risks to users and society, necessitating more effective and scalable content moderation strategies. Current approaches rely on human moderators, supervised classifiers, and large volumes of training data, and often struggle with scalability, subjectivity, and the dynamic nature of harmful content (e.g., violent content, dangerous challenge trends, etc.). To bridge these gaps, we utilize Large Language Models (LLMs) to undertake few-shot dynamic content moderation via in-context learning. Through extensive experiments on multiple LLMs, we demonstrate that our few-shot approaches can outperform existing proprietary baselines (Perspective and OpenAI Moderation) as well as prior state-of-the-art few-shot learning methods, in identifying harm. We also incorporate visual information (video thumbnails) and assess if different multimodal techniques improve model performance. Our results underscore the significant benefits of employing LLM based methods for scalable and dynamic harmful content moderation online.

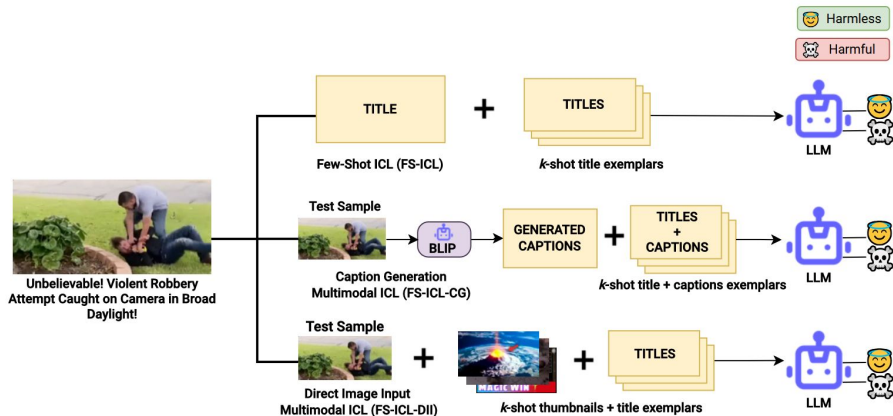


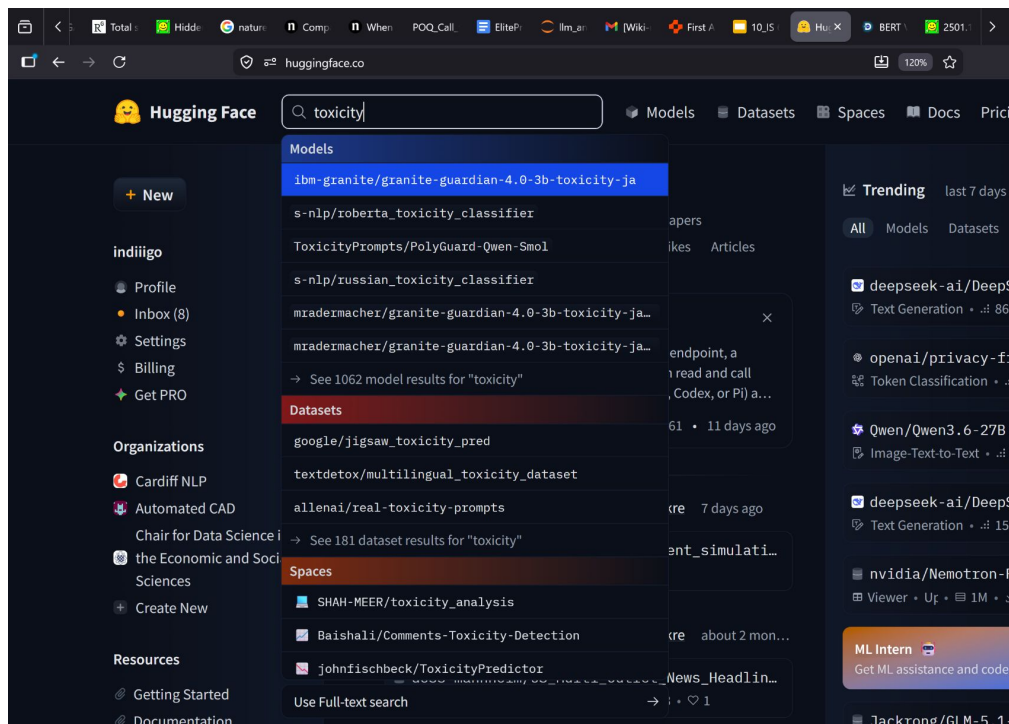
Figure 1: Our overall experimental framework utilizing LLMs for k -shot harmful content classification. The top approach showcases few-shot text-only in-context learning using the video’s title (FS-ICL). The other two methods (FS-ICL-CG) are multimodal and augment the video title information with visual input (e.g. video thumbnail). In FS-ICL-CG we utilize a caption generation module (BLIP) to convert the visual input into text and feed these to a text-only LLM. In FS-ICL-DII we utilize fully multimodal LLMs and feed the visual input to the model directly. The output of each method is a classification of *Harmful/Harmless* for each given test input. Note that by setting $k = 0$ we can obtain zero-shot learning (ZSL) variants for each of the aforementioned approaches.

BERT-style and Generative LLMs have many drawbacks

- Much more **resource intensive** than non deep-learning approaches
 - Pre-training is near impossible for people without GPUs, especially bigger models
 - Bigger models (i.e., more parameters) → more resources
 - Even fine-tuning and inference require access to GPUs → you will need to do this on the BWUniCluster
 - Much higher environmental impact
- **Intransparent:** not really clear what is going on inside the model
 - Deep learning interpretability is an active area of research
 - But often a lack of clarity on what data was used to train models in the pre-training stage
 - Issues with copyright, toxic content
 - Especially problematic for commercial models where there is little info on pre-training data
- Many **societal biases** encoded in these models

Generative LLMs have many drawbacks for data annotation

- LLMs are brittle; small changes in the prompt (even an extra space) can completely change model output
- LLMs are stochastic; they are not deterministic and can give different results for the very same prompt and same hyperparameters
- Often overkill for constructs where you have other classifiers
 - Search for existing models on <https://huggingface.co/>



An overview of issues with LLM context Analysis/Evaluation

LLM-Evaluation Tropes: Perspectives on the Validity of LLM-Evaluations

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Abstract

Large Language Models (LLMs) are increasingly used to evaluate information retrieval (IR) systems, generating relevance judgments traditionally made by human assessors. Recent empirical studies suggest that LLM-based evaluations often align with human judgments, leading some to suggest that human judges may no longer be necessary, while others highlight concerns about judgment reliability, validity, and long-term impact. As IR systems begin incorporating LLM-generated signals, evaluation outcomes risk becoming self-reinforcing, potentially leading to misleading conclusions.

This paper examines scenarios where LLM-evaluators may falsely indicate success, particularly when LLM-based judgments influence both system development and evaluation. We highlight key risks, including bias reinforcement, reproducibility challenges, and inconsistencies in assessment methodologies. To address these concerns, we propose tests to quantify adverse effects, guardrails, and a collaborative framework for constructing reusable test collections that integrate LLM judgments responsibly. By providing perspectives from academia and industry, this work aims to establish best practices for the principled use of LLMs in IR evaluation.

confirmation email (Conference acronym 'XX). ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Large Language Models (LLMs) are increasingly used in the evaluation of information retrieval (IR) systems, generating relevance judgments that were traditionally the domain of human assessors. Given an information need (or topic) and a set of documents, assessors determine the relevance of each document to the topic. A process that forms the foundation of retrieval evaluation. However, due to the vast number of topic-document pairs, traditional assessment relies on pooling methods to identify a subset of documents for judgment. LLMs offer an alternative approach, with the potential to scale relevance assessments far beyond the limits of human annotation. However, Soboroff [64] writes: "Letting the LLM write your truth data handicaps the evaluation by setting that LLM as a ceiling on performance." In this paper we aim for a middle ground by discussing 13 ways in which LLMs can negatively impact the evaluation and how avoid this adverse effect.

Figure 1: LLM-evaluation tropes that can lead to invalid conclusions about evaluation, systems, and the efficacy of human judges that oversee the process. Overarching patterns are circularity $\cup$, Goodhart's law $\heartsuit$, and loss of variety $\cong$.

Evaluation Tropes:

- $\cup$ #1 **Circularity**: Leaking the evaluation signal into the IR system.
- $\cup$ #2 **LLM-Eval as a Ranker**: Using the same approach in the system and the evaluation.
- $\heartsuit$ #3 **LLM Narcissism**: LLMs prefer text from their own model.
- $\cong$ #4 **Loss of Variety of Opinion**: When all judges think alike.

Meta-Evaluation Tropes:

- $\heartsuit$ #5 **Ignored Label Correlation**: When human and LLM judges disagree on relevance labels.
- $\cong$ #6 **Old Systems**: Evaluators need to identify the best systems of the future.
- #7 **LLM Evolution**: LLMs are not static; they can improve or degrade over time.

System Tropes:

- $\heartsuit$ #8 **Test Set Leak**: LLMs trained on test collections create the illusion of quality.
- $\cup$ #9 **Self-Training Collapse**: Concept drift from training LLMs on LLM output.
- $\heartsuit$ #10 **Goodhart's Overfitting**: IR systems that are "trained for the test".
- $\heartsuit$ #11 **Adversarial Threats**: Bad actors want to manipulate the systems and evaluation.

Judge Tropes:

- $\cong$ #12 **Rubber-Stamp Effect**: Lack of critical oversight when humans blindly trust LLM labels.
- #13 **Black-box Labeling**: When relevance is complex, labels may be difficult to interpret.
- $\cup$ #14 **Predictable Secrets**: When human data can be guessed by an LLM.

How can you use Generative LLMs for your projects?

- Generative LLMs are a good option if:
 - You do not have an off-the-shelf classifier for your construct
 - You do not have enough resources to create a training dataset for fine-tuning supervised modelson encoder-only models like BERT
- If your dataset is too large to have everything labeled by Generative LLMs, you can opt for a three-step pipeline:
 - Get some instances annotated by a generative LLM (~1000 per class or more)
 - Conduct a small manual validation (~20-50 per class) of the LLM labels
 - If you find the LLM labels are good, fine-tune a decoder-based classifier on these labels (e.g., BERT, RoBERTa)
 - Else, repeat the generative-LLM labeling with better prompts

Best Practices when using LLMs for Data Annotation

- Clear prompts
- Experiment with different prompting modes
 - Definition of construct (none, detailed, minimal)
 - Examples, i.e., few-shot (think about how to pick examples)
 - Enhancements: Chain-of-thought
- Conduct **multiple runs** rather than one single run
 - Clearer understanding of model reliability

	Toxicity			Sentiment			Rumor stance		
	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT
Definition (yes)	24.20	81.13	73.11	—	—	—	7.27	60.87	61.51
Definition (no)	31.36	82.70	69.45	28.78	53.23	41.20	7.35	61.65	61.83
Explanation (yes)	23.23	81.13	71.22	34.11	55.70	36.21	7.28	60.04	63.72
Explanation (no)	32.25	82.70	71.34	22.96	50.33	46.11	7.37	62.55	59.62
Output type (label)	17.68	85.09	73.99	31.00	57.52	46.39	7.25	67.44	69.87
output type (score)	37.73	78.75	68.56	24.27	49.66	35.91	7.55	53.18	53.44
Length (standard)	31.85	85.35	73.38	27.12	51.45	39.46	7.57	61.62	63.38
Length (concise)	23.67	78.48	69.17	30.53	55.56	42.97	7.07	61.03	59.95

Table 5: LLM percentage accuracy for different prompt designs.

[Atreja et al., “Prompt Design Matters for Computational Social Science Tasks but in Unpredictable Ways”](#)

Best Practices when using LLMs for Data Annotation

- Reproducibility is vital!
 - Use open-source LLMs as much as possible => less risk of sudden model retirement
 - Always document prompts and model hyperparameters
- Try different LLMs
- See if different LLMs have high agreement => one sign of data annotation validity
- Validate!
 - Manual validation
 - Mixed-methods validation, e.g., TF-IDF of tokens in different classes

Group Activity: Use an LLM for Data Annotation

- Use the web version of an LLM, e.g., ChatGPT, Claude, Gemini
- Each member of the group should pick one model
- Prompt the model to annotate one or two instances from the social media data you collected with the construct that you're interested in: Hype, Sentiment towards ICE, etc
- Experiment with different prompts, especially with examples, Chain-of-thought, definitions, etc
- How well does the LLM label the construct?

Summary and Next Week

- This session we learned:
 - Topic Modeling and other unsupervised methods
 - A Primer on BERT
 - A Primer on LLMs
 - How you can use LLMs in your project
- Next week:
 - Network analysis or causal inference?
- Midterm pitches will take place in the last week of the semester during the exercise slots (27.05, 28.05)
 - 10 minutes per team
 - Each exercise will have 3 presentations
 - The penultimate week will be for project consultations (lecture and exercises)

Questions?

Additional Resources

- Practical Tutorial on BERT:
<https://colab.research.google.com/drive/1YtTqwkwaqV2n56NC8xerflt95Cjud4NE?usp=sharing>
- Uni Mannheim course on Advanced Text Analytics:
<https://www.uni-mannheim.de/dws/teaching/course-details/courses-for-master-candidates/ie-696-advanced-methods-in-text-analytics/>
- Extensive set of tutorials on BERT and LLMs:
<https://aiforhumanists.com/workshops/>